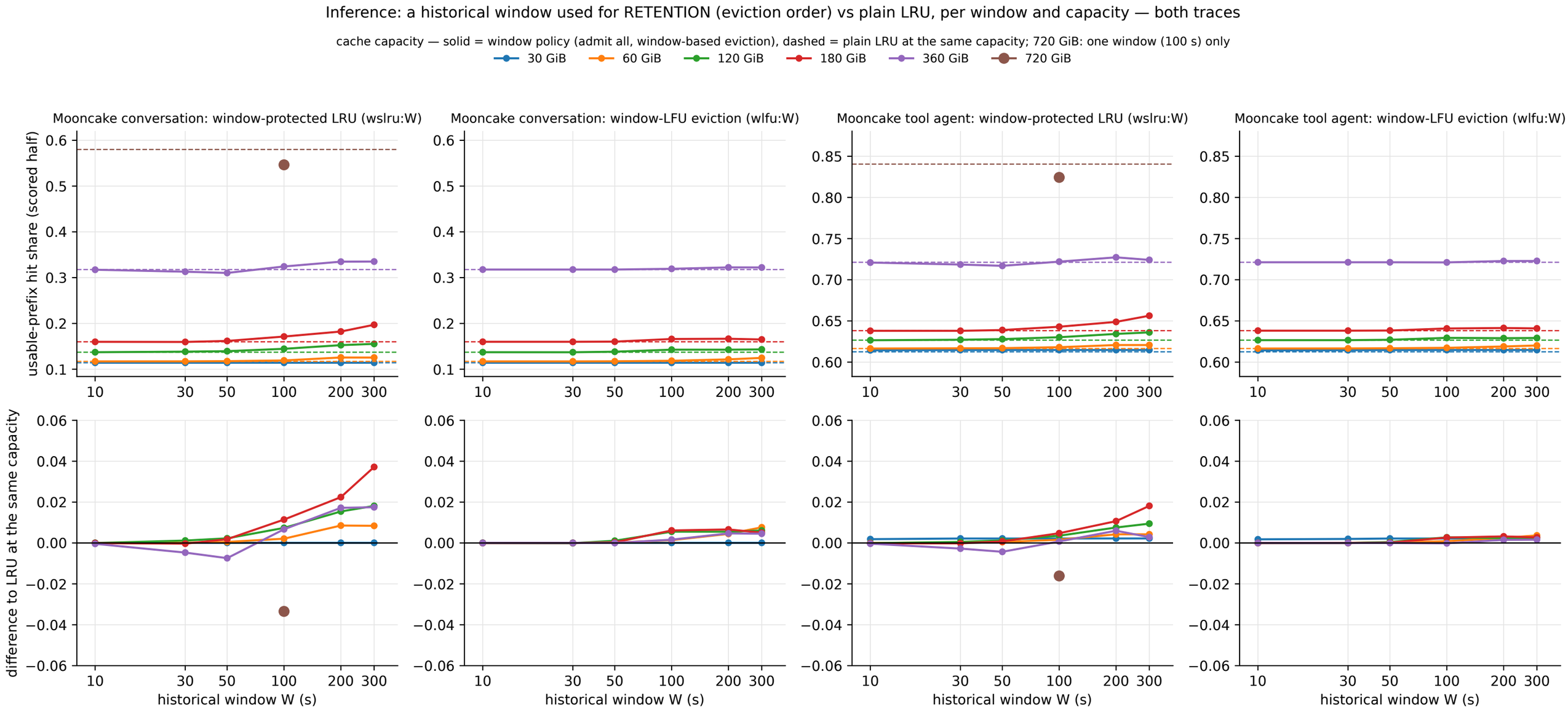

KV-cache admission replay (kv_admission_sim.py) of the Mooncake conversation and tool-agent traces; one host cache of 8B blocks (67.1 MB); the first half of each trace warms the cache, the second half is scored. Usable-prefix hit share = share of the scored block lookups with a reuse that hit AND whose whole preceding prefix in the request also hit (a hit the engine can use). wslru:W = LRU with a protected segment for blocks with ≥ 2 accesses inside the last W s (protection expires after W s idle); wlfu:W = among the K least recently used blocks evict the one with the fewest accesses in the last W s; both admit every block (like LRU). Historical windows only; no future information is used online.